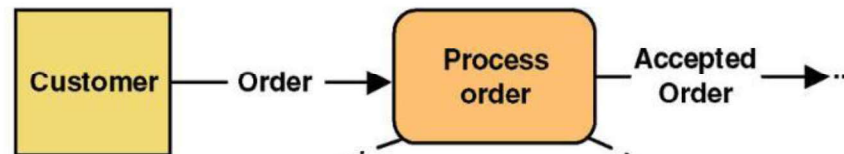


Chapter 7

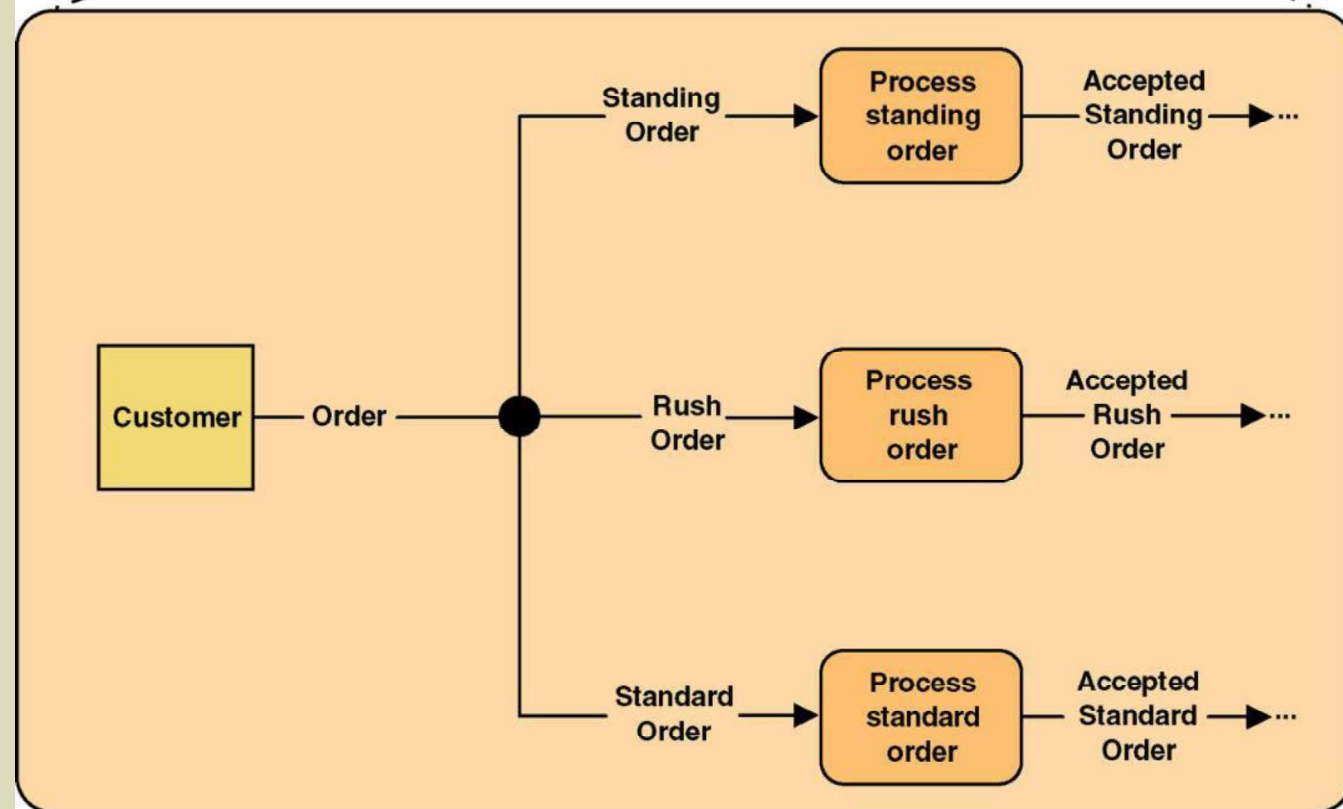
Process Modeling

Composite and Elementary Data Flows

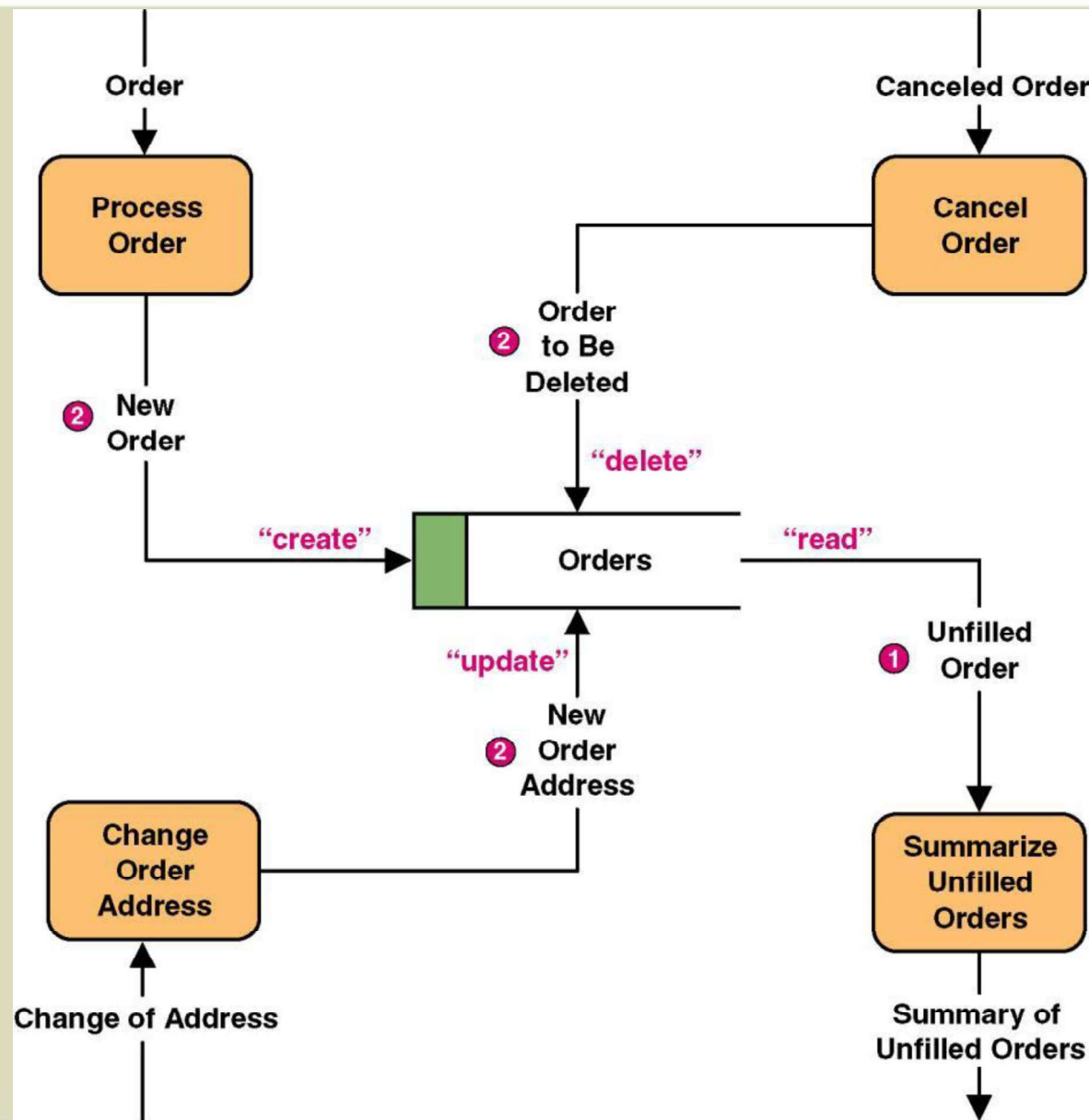
(a) High-Level DFD



(b) More Detailed DFD



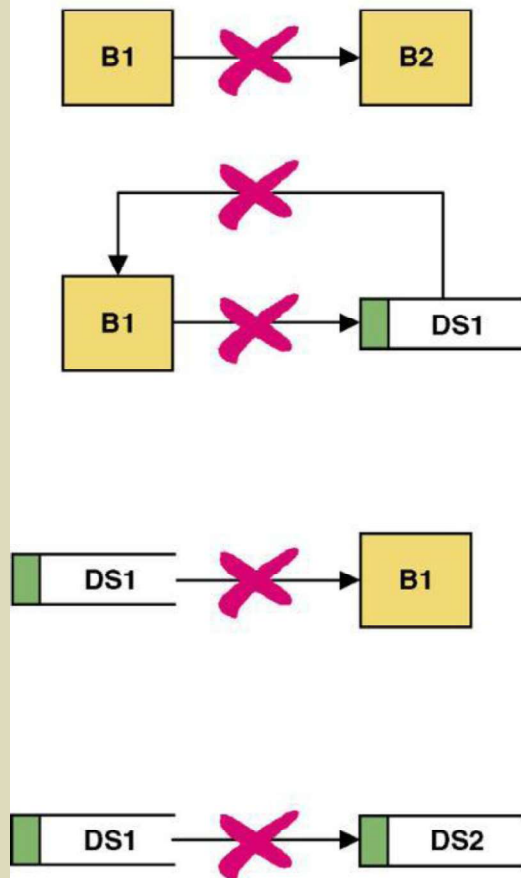
Data Flows To and From Data Stores



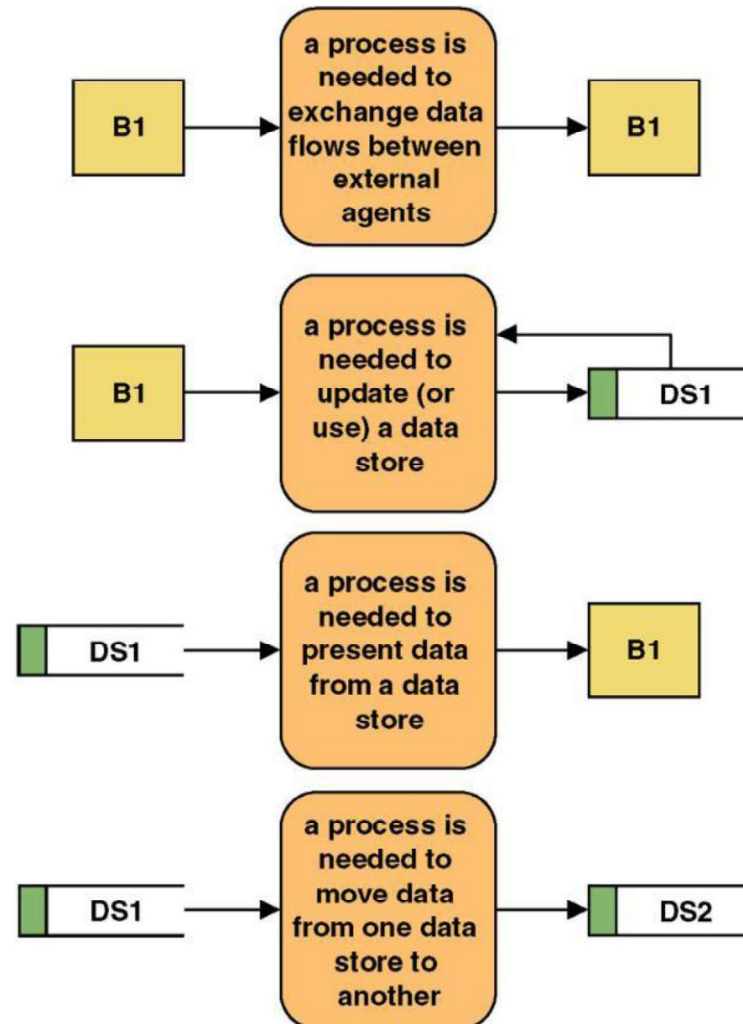
Illegal Data Flows



Illegal data flows



Corrected data flows



Diverging and Converging Data Flows

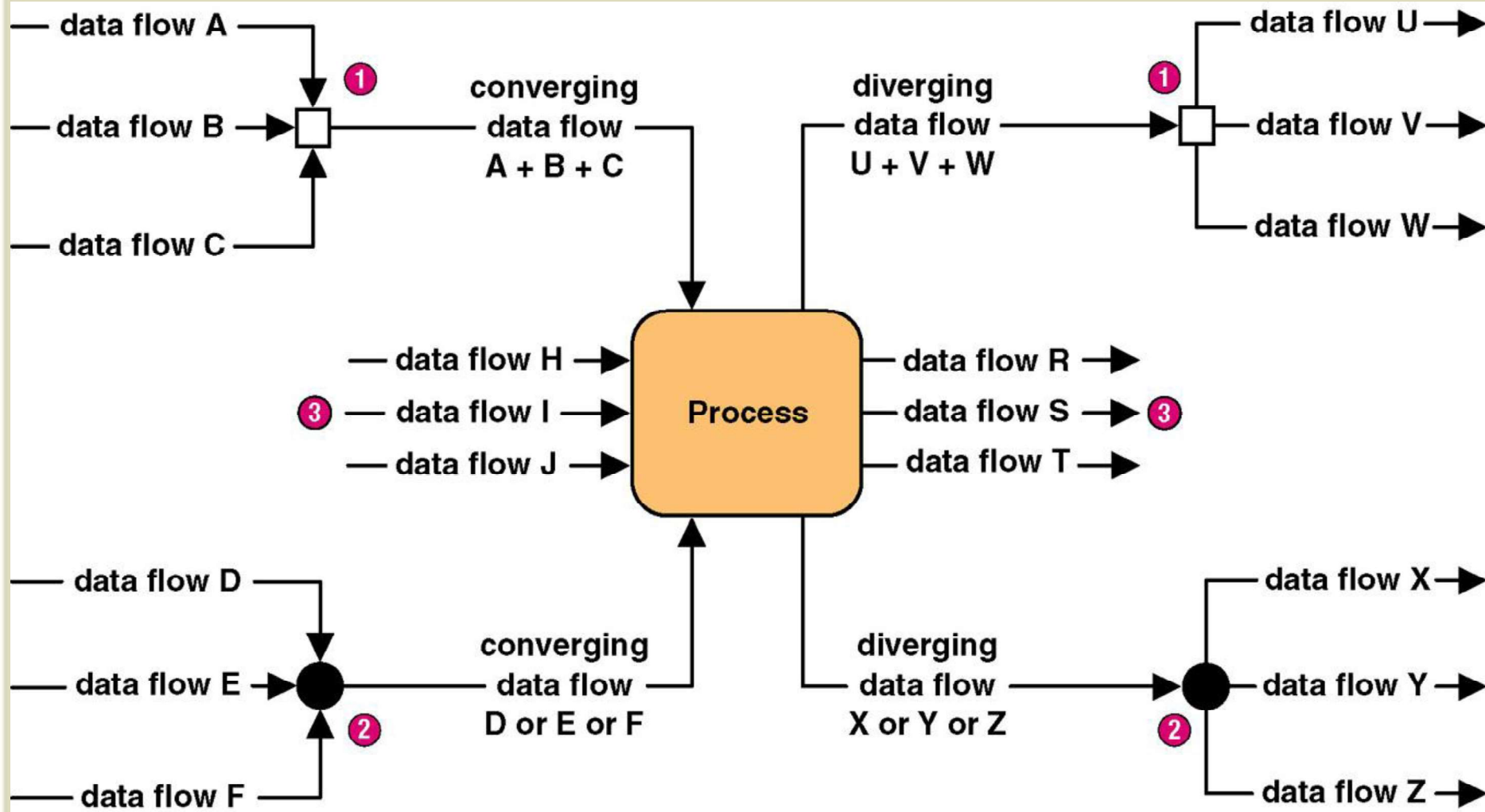
Diverging data flow – a data flow that splits into multiple data flows.

- Indicates data that starts out naturally as one flow, but is routed to different destinations.
- Also useful to indicate multiple copies of the same output going to different destinations.

Converging data flow – the merger of multiple data flows into a single packet.

- Indicates data from multiple sources that can (must) come together as a single packet for subsequent processing.

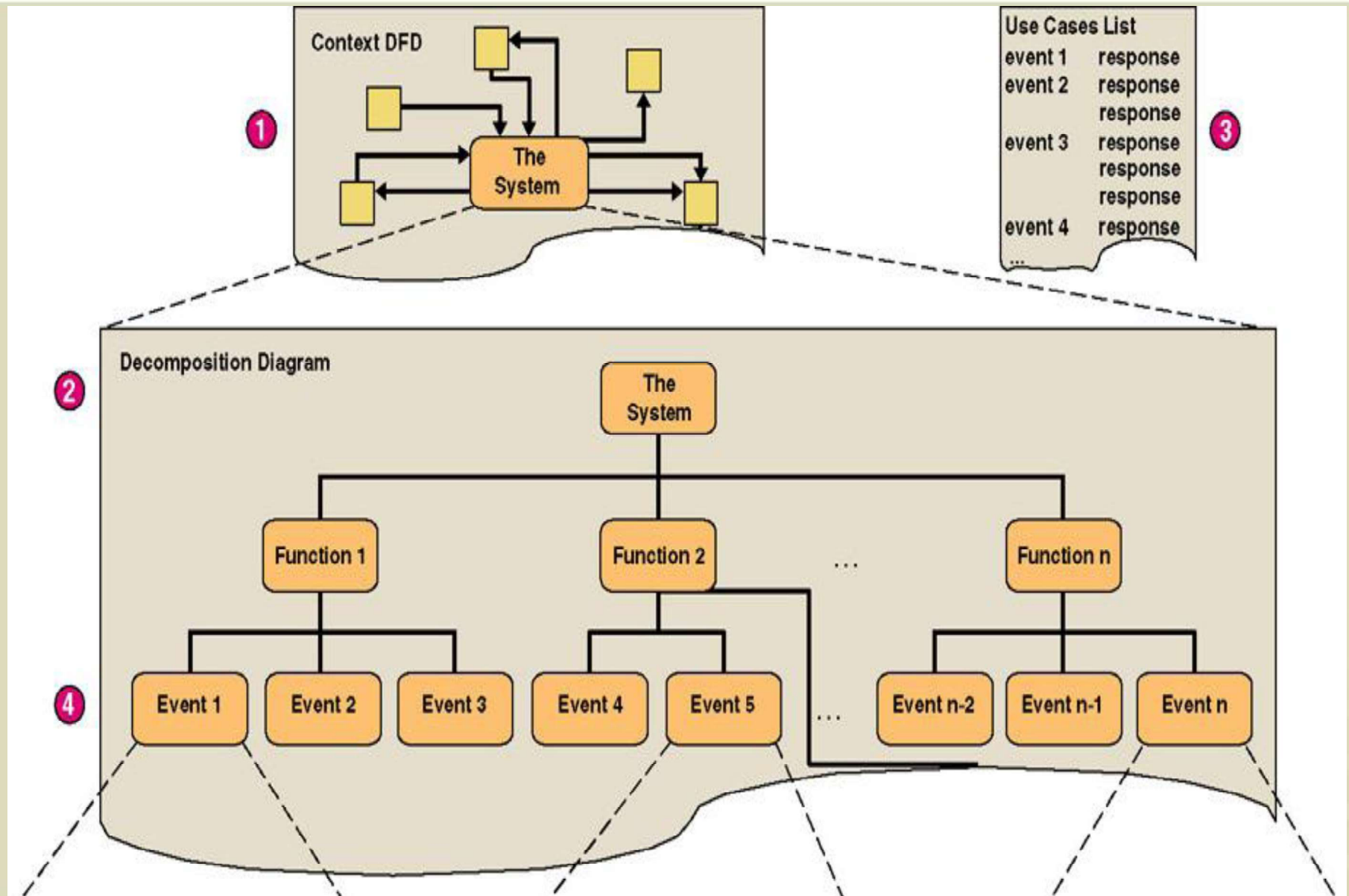
Diverging and Converging Data Flows



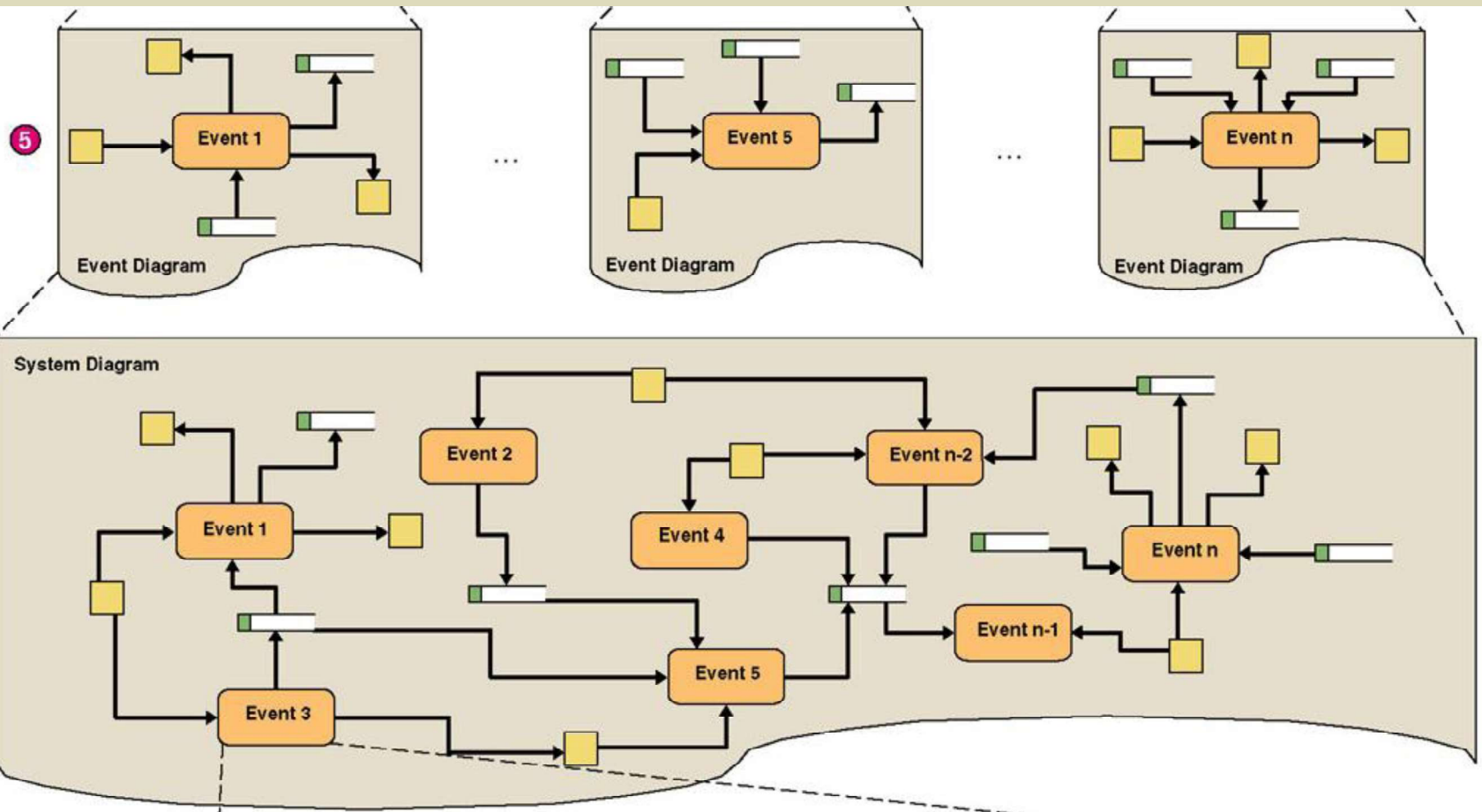
Modern Structured Analysis

1. Draw a context DFD to establish initial project scope.
2. Draw a functional decomposition diagram to partition the system into subsystems.
3. Create an event-response or use-case list for the system to define events for which the system must have a response.
4. Draw an event DFD (or event handler) for each event.
5. Merge event DFDs into a system diagram (or, for larger systems, subsystem diagrams).

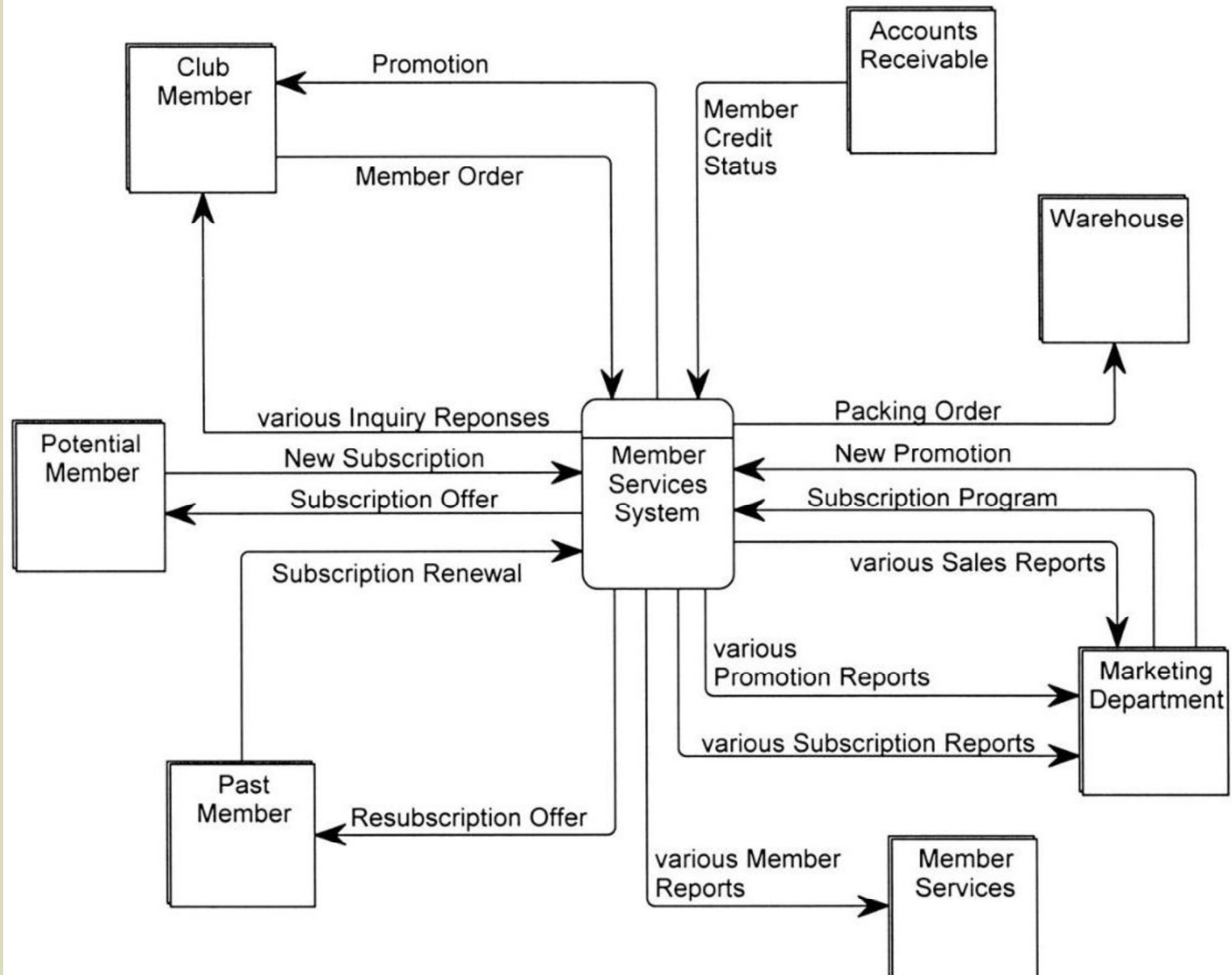
Structured Analysis Diagram Progression



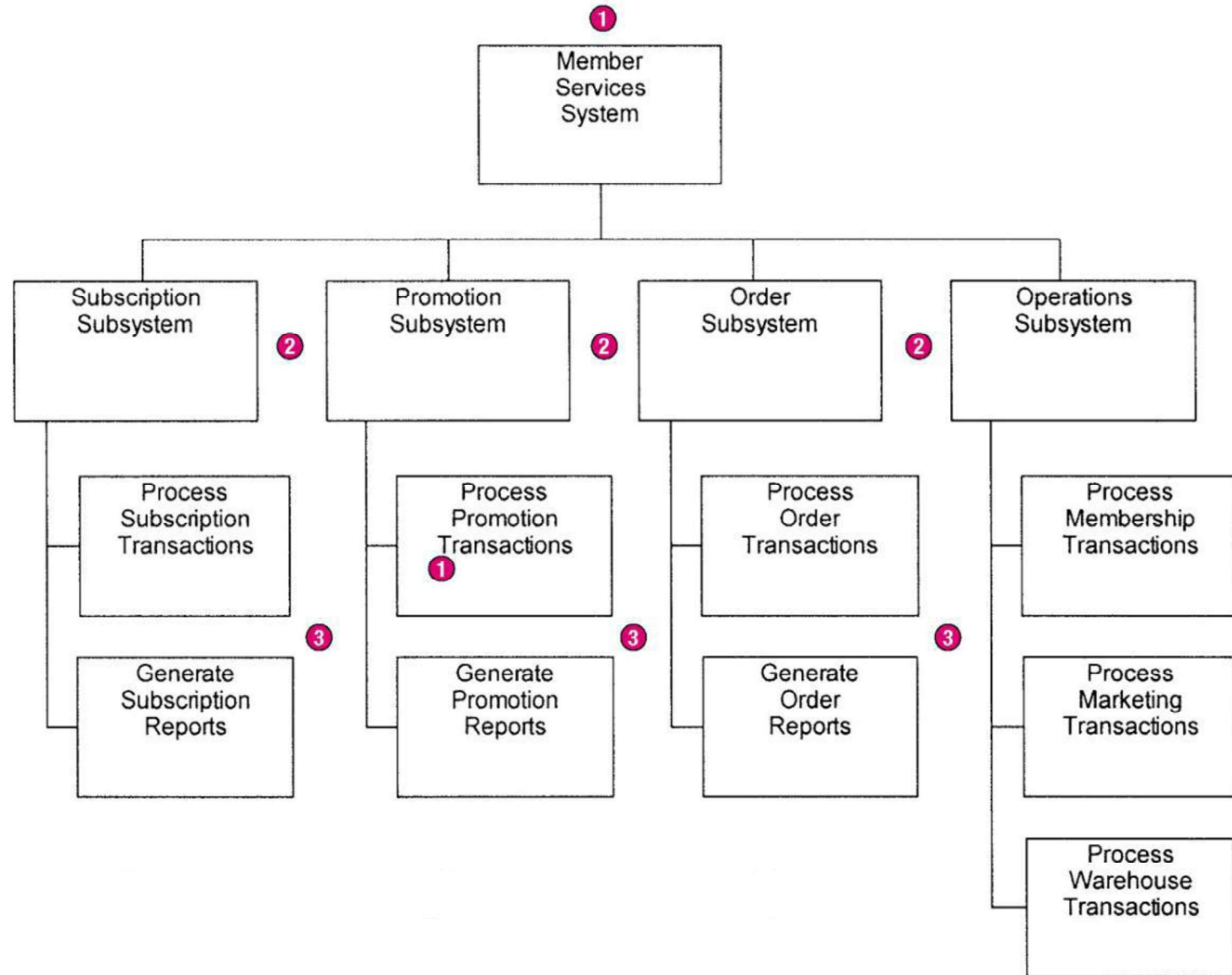
Structured Analysis Diagram Progression



SoundStage Context DFD



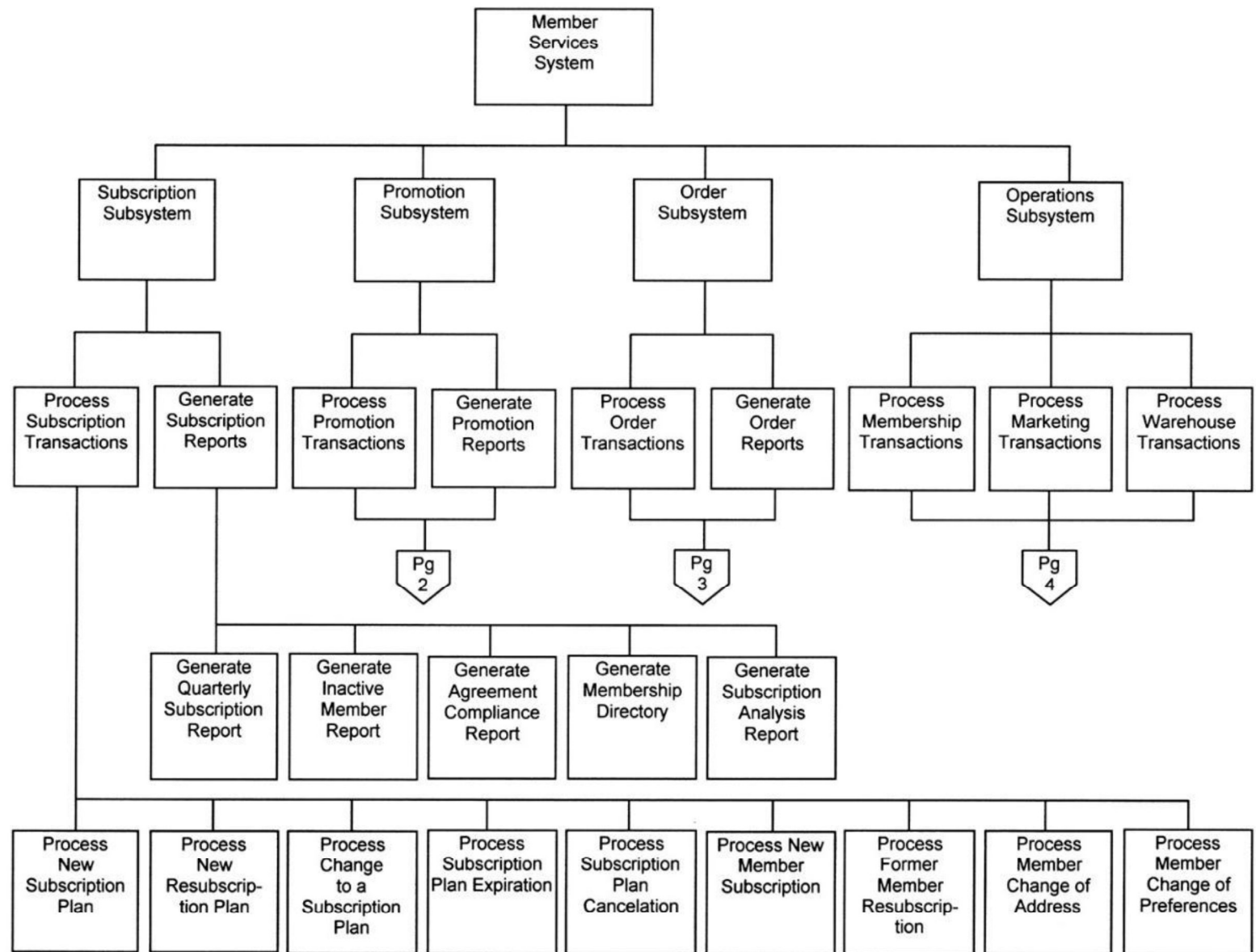
SoundStage Functional Decomposition Diagram



Use Case List

Actor	Event (or Use Case)	Trigger	Responses
Marketing	Establishes a new membership subscription plan to entice new members.	NEW MEMBER SUBSCRIPTION PROGRAM	Generate SUBSCRIPTION PLAN CONFIRMATION. Create AGREEMENT in the database.
Marketing	Establishes a new membership resubscription plan to lure back former members.	PAST MEMBER RESUBSCRIPTION PROGRAM	Generate SUBSCRIPTION PLAN CONFIRMATION. Create AGREEMENT in the database.
Marketing	Changes a subscription plan for current members (e.g., extending the fulfillment period).	SUBSCRIPTION PLAN CHANGE	Generate AGREEMENT CHANGE CONFIRMATION. Update AGREEMENT in the database.
(time)	A subscription plan expires.	(current date)	Generate AGREEMENT CHANGE CONFIRMATION. Logically Delete (void) AGREEMENT in the database.
Marketing	Cancels a subscription plan before its planned expiration date.	SUBSCRIPTION PLAN CANCELLATION	Generate AGREEMENT CHANGE CONFIRMATION. Logically Delete (void) AGREEMENT in the database.
Member	Joins the club by subscribing. ("Take any 12 CDs for one penny and agree to buy 4 more at regular prices within two years.")	NEW SUBSCRIPTION	Generate MEMBER DIRECTORY UPDATE CONFIRMATION. Create MEMBER in the database. Create first MEMBER ORDER and MEMBER ORDERED PRODUCTS in the database.
Member	Changes address (including e-mail and privacy code).	CHANGE OF ADDRESS	Generate MEMBER DIRECTORY UPDATE CONFIRMATION. Update MEMBER in the database.
Accounts Receivable	Changes member's credit status.	CHANGE OF CREDIT STATUS	Generate CREDIT DIRECTORY UPDATE CONFIRMATION. Update MEMBER in the database.
(time)	90 days after Marketing decides to no longer sell a product.	(current date)	Generate CATALOG CHANGE CONFIRMATION. Logically Delete (deactivate) PRODUCT in the database.

Event Decomposition Diagram (partial)



System DFD

